

AUTOMATED CI/CD PIPELINE

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Automated Cloud Deployment on AWS

IBM presentation

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- DevOps
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THE PROBLEM

Deploying applications manually is slow, repetitive and prone to mistakes. For every small code change, developers usually have to connect to the server, copy files, restart the application and verify that everything still works.

As projects grow, this process becomes unreliable and wastes valuable development time.

Traditional Deployment

- SSH into server
- Copy updated files
- Restart application
- Verify deployment

Challenges

- Human errors
- Downtime
- Slow releases
- Difficult to scale

- DevOps
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THE SOLUTION

- ✓ Fully Automated
- ✓ No Manual Deployment
- ✓ Faster Releases
- ✓ Docker Containers
- ✓ CI/CD Pipeline
- ✓ AWS Managed Services

This project automates the complete deployment process using AWS DevOps services.

A simple Git push automatically builds a Docker image, stores it in Amazon ECR and deploys the latest version to Amazon ECS running on EC2.

WHAT IS DOCKER?



docker

Without Docker

Different operating systems and software versions often cause applications to behave differently on every machine.
This leads to deployment failures.

With Docker

Docker packages the application together with all required dependencies inside a container.
The same container runs consistently on every environment.

Why It Matters

Build once.
Run anywhere.
Develop faster with predictable deployments.

AWS SERVICES USED

CodePipeline

Automates the entire CI/CD workflow.

Amazon ECR

Stores versioned Docker images.

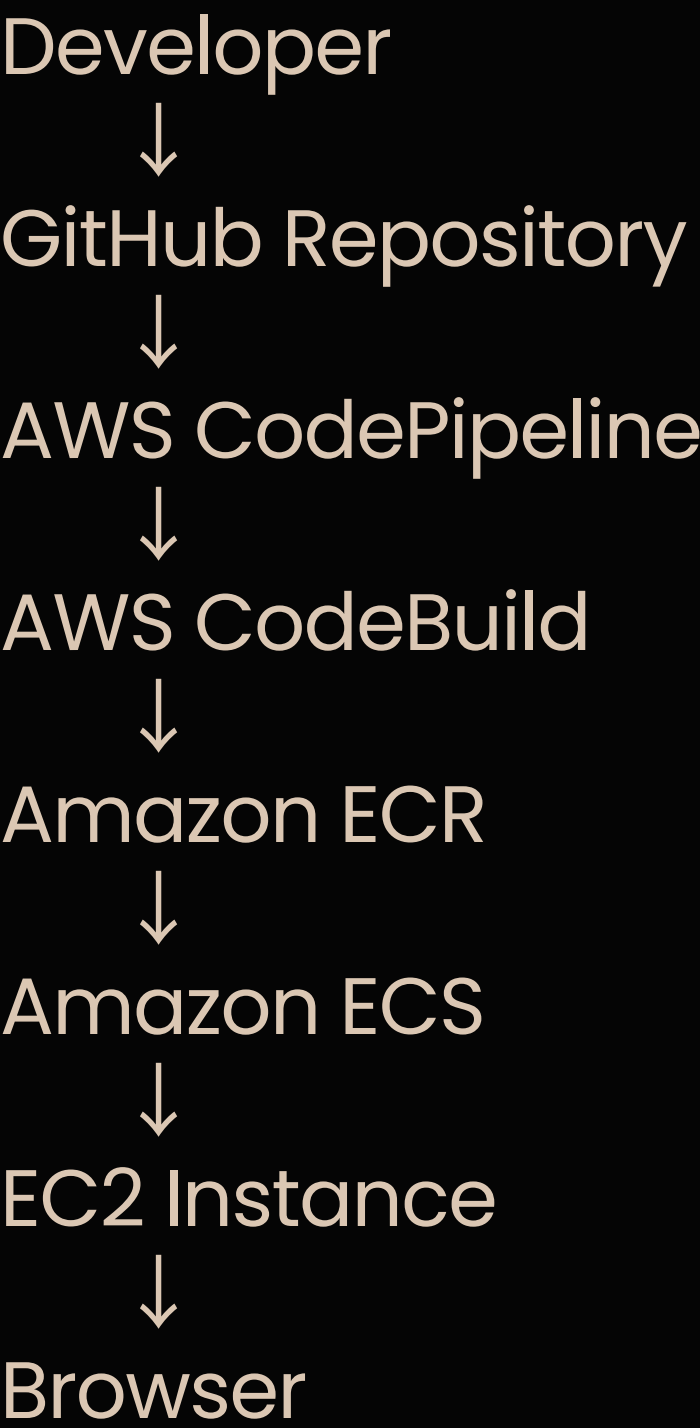
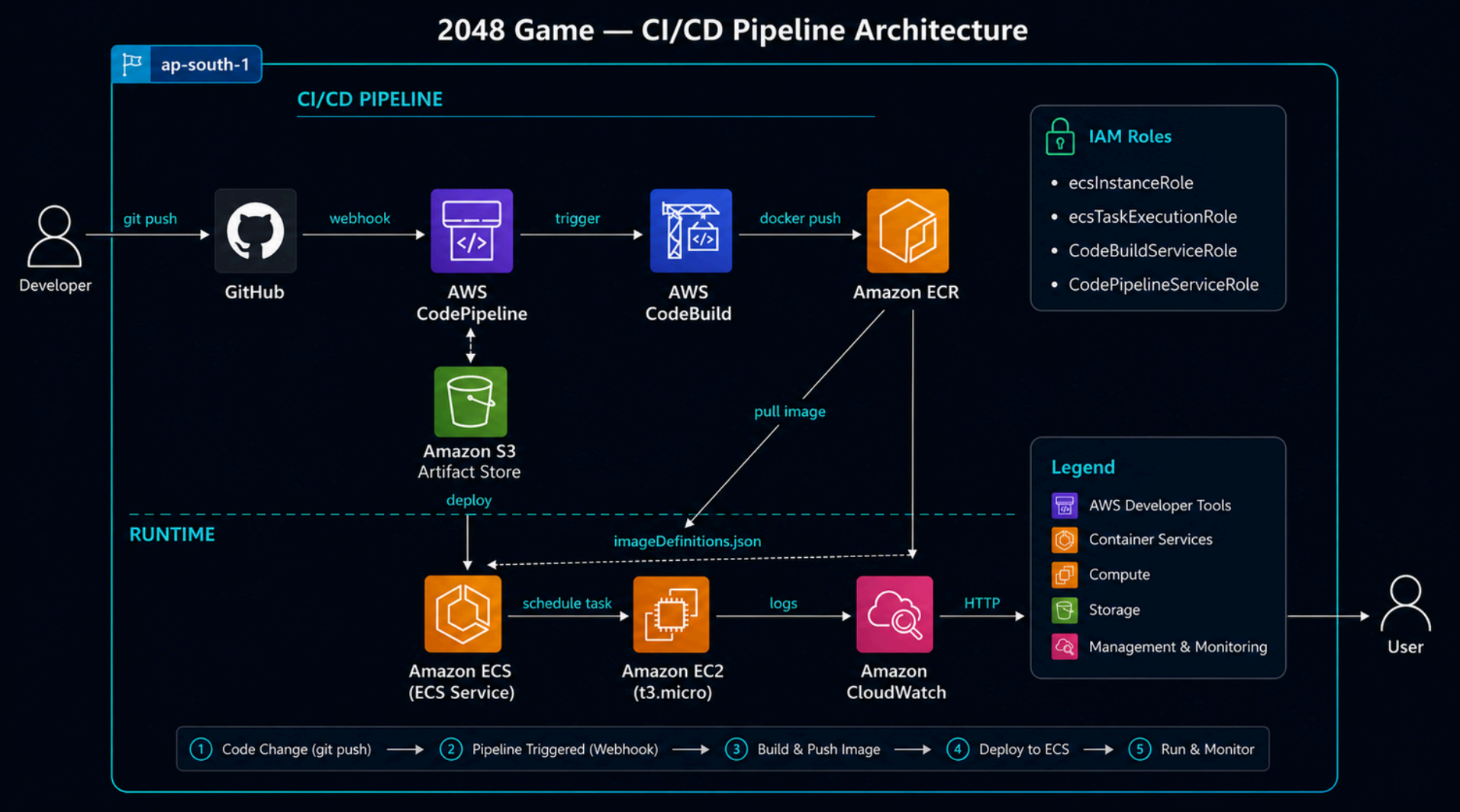
CodeBuild

Builds Docker images automatically.

Amazon ECS

Deploys and manages containers.

ARCHITECTURE



LIVE WORKFLOW

**Let's move from the
architecture
to the actual deployment.**

I'll now demonstrate:

- ✓ **The live application**
- ✓ **The GitHub repository**
- ✓ **The CI/CD pipeline execution**
- ✓ **Automatic deployment on
AWS**

THANK
YOU. *for Listening*